VERIFICATION OF NORTON'S THEOREM

AIM:-

To Verify Norton's theorem practically and theoretically for the given DC circuit.

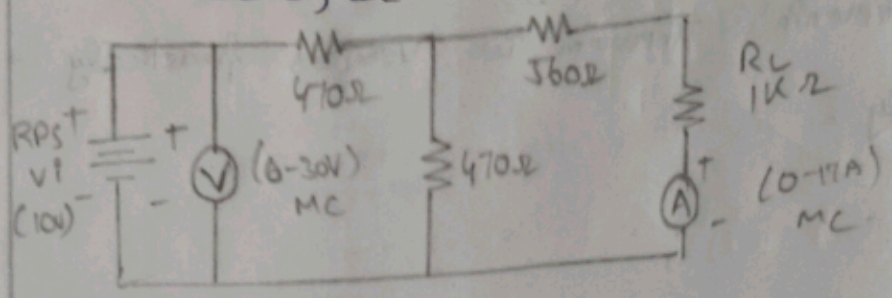
Apparatus required :-

SL.NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated power Supply (RPS)	(0-30V)	1
2.	Voltmeter	(0-30V) MC	1
3.	Ammeter	(0-10)mA MC	1
4.	Resistor	470 Ω	2
		560 Ω	1
		1K Ω	1
5.	Bread board	---	1
6	multimeter	---	1

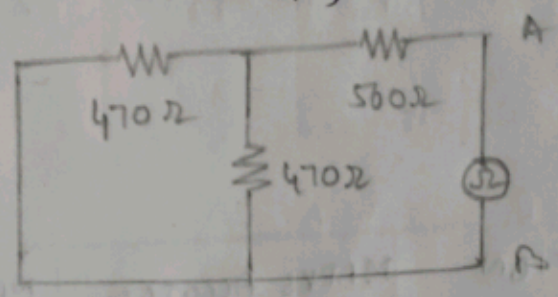
CIRCUIT DIAGRAM:

VERIFICATION OF NORTON'S THEOREM:

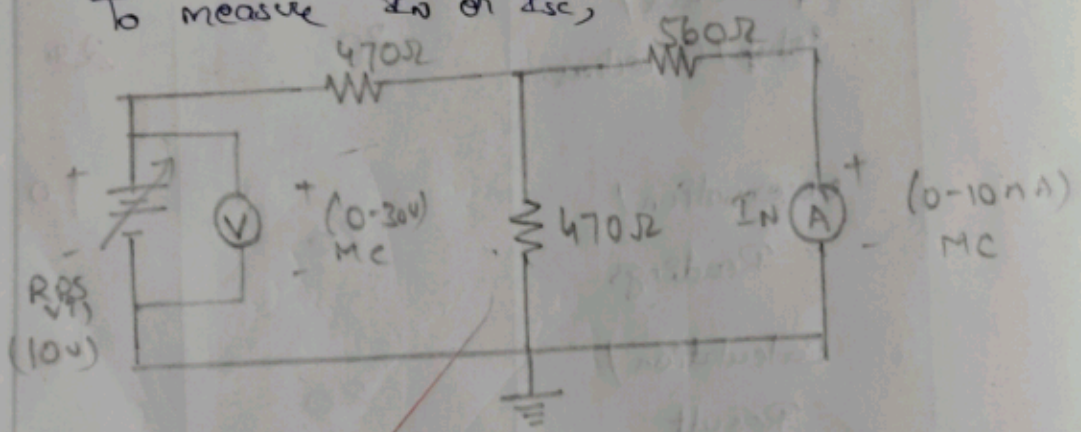
To measure, I_L.



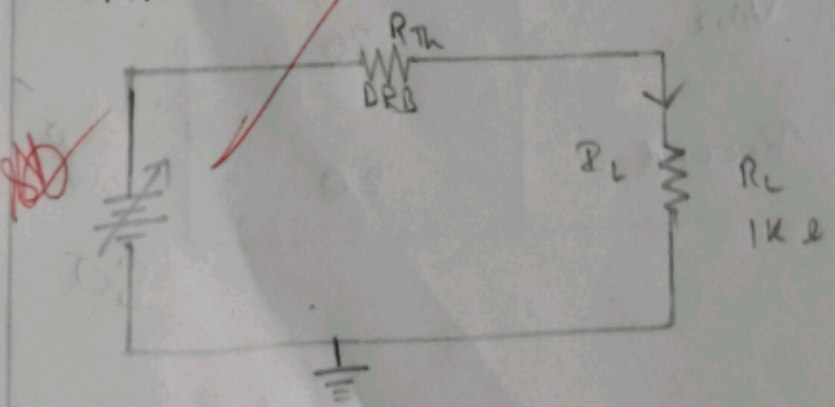
To measure RN,



To measure I_N or I_{sc},



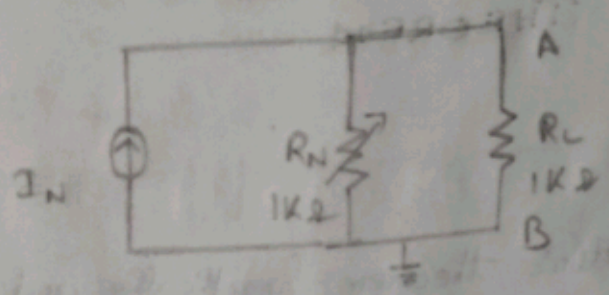
THEVENIN'S EQUIVALENT CIRCUIT:



PROCEDURE :-

1. Make the connections as per the circuit diagram.
2. Vary the RPS and set an Input voltage of 10V.
3. Note down the voltmeter readings (V_i) and ammeter reading (I_L) in
4. Switch off the supply and make connections for circuit 2.
5. Measure the Thevenin's resistance R_{Th} = Norton's R_N.
6. Switch off the supply and make connections for circuit 3.
7. Set an input voltage of 10V in the RPS and note down the voltmeter readings V_i and V_{Th} (= V_{oc})
8. Switch off the supply and make connections.
9. Set an input voltage of 10V in the RPS and note down the voltmeter reading V_i and Ammeter readings I_N (= I_{sc}).
10. Draw the Thevenin's equivalent circuit and Norton's equivalent circuit.

NEWTON'S EQUIVALENT CIRCUIT



To measure I_L,

V_i (volts)	I_L (amps)
10V	2.785mA.

To measure R_N,

V_i (volts)	R_N (Ω)
10V	795Ω

To measure I_N or I_{sc}

V_i (volts)	I_N (amps)
10V	6.289mA.

not verified

11. Calculate the I_L values using the formula

Thevenin's theorem

$$I_L = V_{Th} / (R_{Th} + R_L)$$

Norton's Theorem

$$I_L = I_N * R_N / (R_N + R_L)$$

12. Theoretically verify the Norton's theorem.

MODEL CALCULATION :-

Practical value of I_L (from tabulation 1) = 2.3mA

Verification of Norton's theorem

$$I_L = I_N * R_N / (R_N + R_L) = 2.43mA.$$

Theoretical calculation of I_L, I_N and R_{Th} (R_N) for the given circuit.

CALCULATION:-

Applying KVL,

$$10 = 470(I) + 470(I)$$

$$10 = I(470+470)$$

$$10 = I(940)$$

$$I = \frac{10}{940} = \frac{1}{94}$$

$$I = 0.010638A$$

$$V_N = I \times R_1$$

$$= (0.0106)(470)$$

$$V_N = 4.98V$$

To find R_N

470Ω and 470Ω are parallel

$$= \frac{470 \times 470}{470+470} = \frac{470}{2} = 235\Omega.$$

235Ω and 560Ω in series

$$= 235+560 = 795\Omega$$

$$R_N = 795\Omega.$$

$$I_N = \frac{V_N}{R_N} = \frac{4.98}{795}$$

$$= 6.26 \times 10^{-3} A$$

$$I_N = 6.26mA.$$

Result :-

Thus Thevenin's and Norton's theorem is verified practically and theoretically.

80

10/12/2

MARKS SPLIT UP :-

SL.NO	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1	Experimental setup / Connections	20	20
2	observations / Readings	20	20
3	Calculations / Results	20	10
4	Viva	20	20
5	Record.	20.	20